

THIS WEEK'S TOPIC

The Ideal Gas Equation of State

Ideal-Gas Equation of State

$$PV = RT$$

$$\left\{ \begin{array}{l} P = \text{absolute pressure (kPa)} \\ v = \text{specific volume (m}^3/\text{kg)} \\ R = \text{gas constant (kPa}\cdot\text{m}^3/(\text{kg}\cdot\text{K))} \\ T = \text{temperature (K)} \end{array} \right.$$

Mass

$$mR = (MN)R = NR_u \longrightarrow PV = NR_u T$$

Molar mass (M) x Mole number (N)

$$V = m v \longrightarrow PV = mRT$$

$$R = \frac{R_u}{M} \quad (\text{kJ/kg}\cdot\text{K or kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})$$

$$\left\{ \begin{array}{l} R = \text{gas constant} \\ M = \text{molar mass (kg/kmol)} \\ R_u = \text{universal gas constant} \end{array} \right.$$

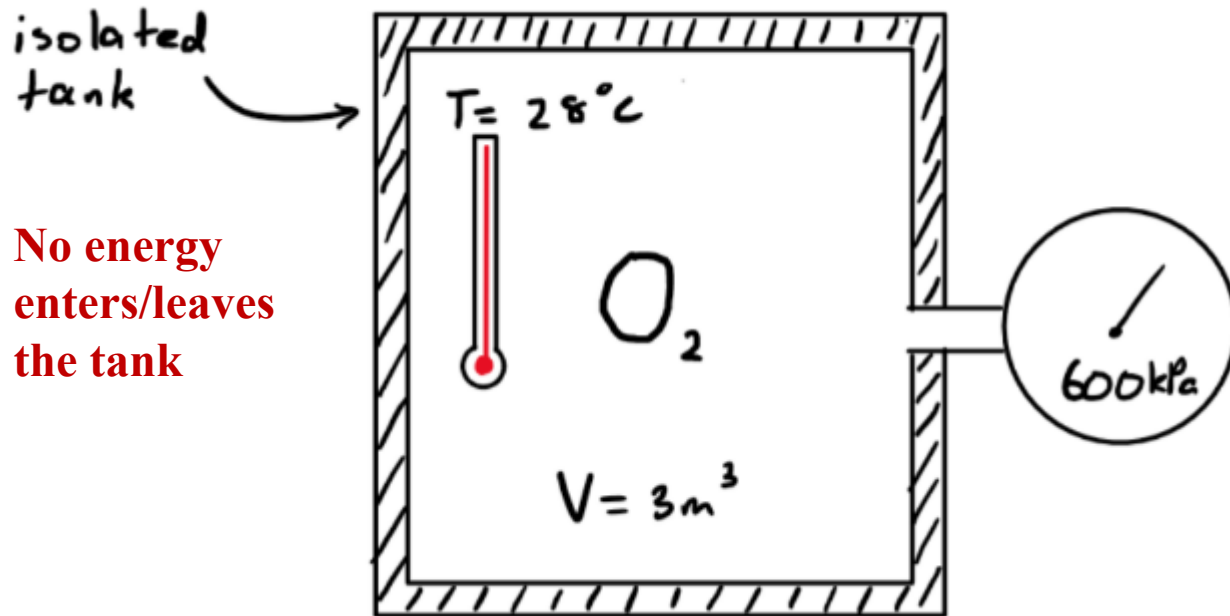
Ideal gas equation at two states for a fixed mass:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Substance	R (kJ/kg·K)
Air	0.2870
Helium	2.0769
Argon	0.2081
Nitrogen	0.2968

Q9. The pressure gauge on a 3m^3 oxygen tank reads 600 kPa. **Determine the amount of oxygen** in the tank if the temperature is 28°C and the atmospheric pressure is 97 kPa.

Analysis:



The Ideal Gas Equation of State

Gas constant for oxygen
check Table A-1 in textbook

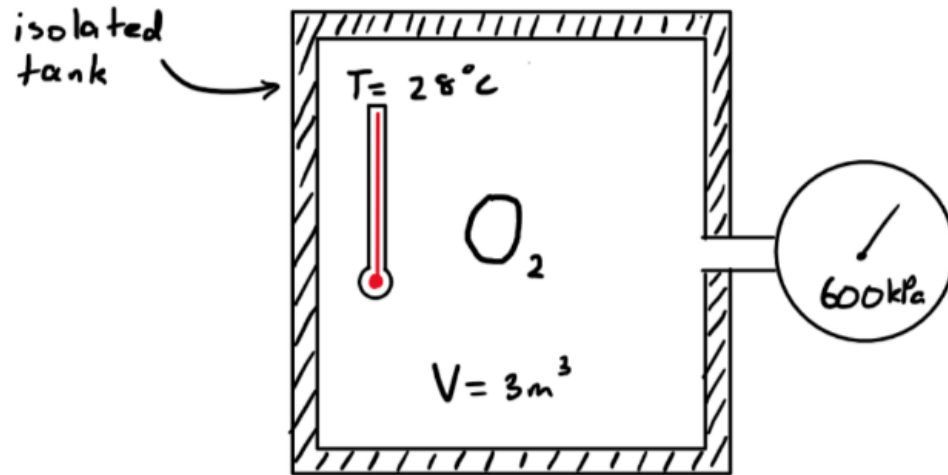
$$Pv = RT$$

Absolute Pressure

$$v = mV \longrightarrow PV = mRT$$

Q9. The pressure gauge on a 3m^3 oxygen tank reads 600 kPa. **Determine the amount of oxygen** in the tank if the temperature is 28°C and the atmospheric pressure is 97 kPa.

Solution:



$$Pv = RT$$

$$V = mU \longrightarrow PV = mRT$$

$$P_{\text{abs}} = P_{\text{atm}} + P_{\text{gage}} = 97\text{kPa} + 600\text{ kPa} = 697\text{ kPa}$$

$$= \mathbf{697,000\text{ Pa}}$$

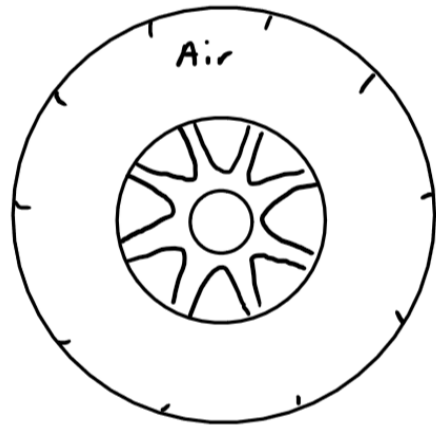
Then, find gas constant: $R_{\text{O}_2} = 0.2598\text{ (kJ/kg} \cdot \text{K)}$

Apply ideal gas state equation to find mass:

$$m = \frac{PV}{RT} = \frac{697\text{kPa} \times 3\text{m}^3}{0.2598\text{ kJ/kg} \cdot \text{K} \times 301.15\text{K}} = \mathbf{26.7259\text{ kg}}$$

Q10. An automobile tire with a volume of 0.5 m^3 is inflated to a gage pressure of 250 kPa. Calculate the mass of air in the tire if the temperature is 25°C using the ideal-gas equation of state.

Analysis:



$$\begin{aligned} P_{\text{gage}} &= 250 \text{ kPa} \\ V &= 0.5 \text{ m}^3 \\ T &= 298.15 \text{ K} \end{aligned}$$

Consider the tire as an **isolated system**, then no energy is transferred into/out of the tire.

Gas constant for **air**
check Table A-1 in textbook

$$Pv = RT$$

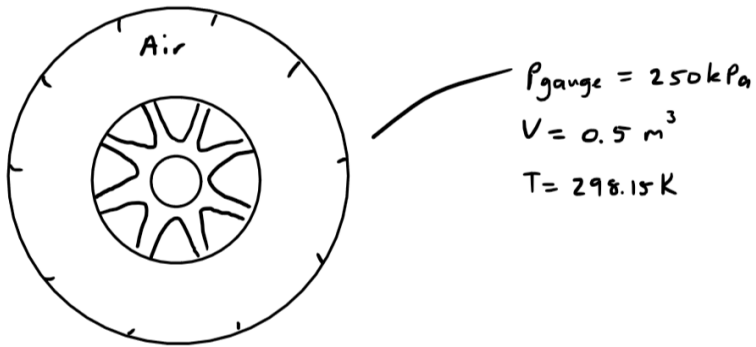
Absolute Pressure

P_{atm} is not given, so the standard atmospheric pressure can be used:

$$P_{\text{atm}} = 101.325 \text{ kPa.}$$

Q10. An automobile tire with a volume of 0.5 m^3 is inflated to a gage pressure of 250 kPa. Calculate the mass of air in the tire if the temperature is 25°C using the ideal-gas equation of state.

Solution:



$$P_{\text{abs}} = P_{\text{atm}} + P_{\text{gage}} = 250 \text{ kPa} + 101.325 \text{ kPa} = 351.3250 \text{ kPa}$$

$$= \mathbf{351,325 \text{ Pa}}$$

Then, find gas constant: $R_{\text{air}} = 0.2870 \text{ (kJ/kg} \cdot \text{K)}$

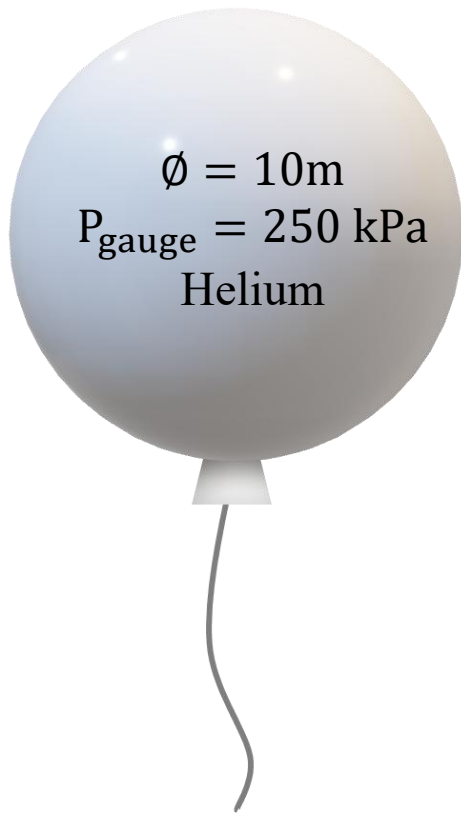
Apply ideal gas state equation to find mass:

$$m = \frac{PV}{RT} = \frac{351.3250 \text{ kPa} \times 0.5 \text{ m}^3}{0.2870 \text{ kJ/kg} \cdot \text{K} \times 298.15 \text{ K}} = \mathbf{2.0529 \text{ kg}}$$

$$U = mU \longrightarrow PV = mRT$$

Q11. A **spherical** balloon with a diameter of 10 m is filled with helium at 25°C and a gauge pressure of 250 kPa. **Determine the mole number and the mass of the helium in the balloon.**

Analysis:



Consider the balloon as a perfect sphere,
calculate the volume by: $V = \frac{4}{3} \pi r^3$

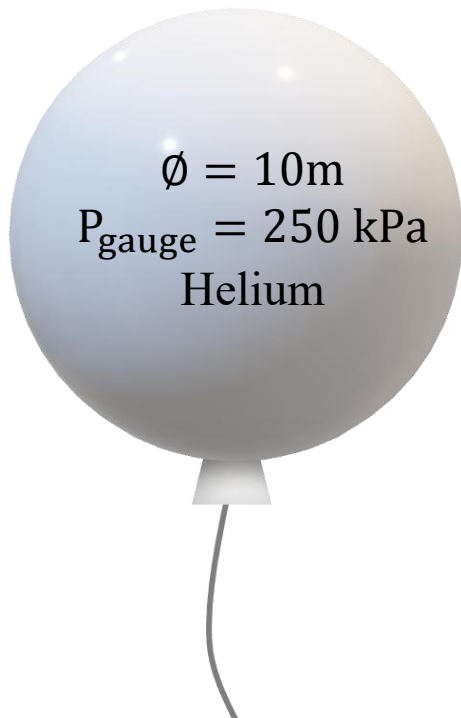
Apply the ideal gas equation to find the mole number:

Mass

$$\overset{\text{Mass}}{\boxed{m}} R = \underbrace{(MN)}_{\text{Molar mass (M) x Mole number (N)}} R = NR_u \longrightarrow PV = NR_u T$$

Q11. A **spherical** balloon with a diameter of 10 m is filled with helium at 25°C and a gauge pressure of 250 kPa. **Determine the mole number and the mass of the helium in the balloon.**

Solution -1:



$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \times 5^3 = 523.5988 \text{ m}^3$$

$$P_{\text{abs}} = P_{\text{atm}} + P_{\text{gage}} = 250\text{kPa} + 101.325 \text{ kPa} = 351.3250 \text{ kPa}$$

$$= \mathbf{351,325 \text{ Pa}}$$

Then, find gas constant: $R_{\text{He}} = 2.0769 \left(\frac{\text{kJ}}{\text{kg}} \cdot \text{K} \right)$ and

mole mass $M_{\text{He}} = 4.003 \text{ Kg/kmol}$

Apply ideal gas state equation to find mass:

$$m = \frac{PV}{RT} = \frac{351.3250 \text{ kPa} \times 523.988 \text{ m}^3}{2.0769 \text{ kJ/kg} \cdot \text{K} \times 298.15\text{K}} = \mathbf{297.0690 \text{ kg}}$$

$$\text{Then } N = \frac{m}{M} = \frac{297.0690 \text{ kg}}{4.003} = \mathbf{74.21 \text{ kmol}}$$

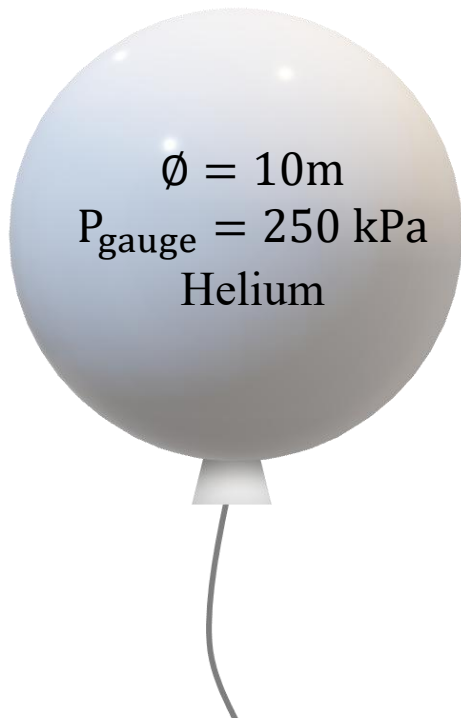
Mass

$$mR = (MN)R = NR_u \longrightarrow PV = NR_uT$$

Molar mass (M) x Mole number (N)

Q11. A **spherical** balloon with a diameter of 10 m is filled with helium at 25°C and a gauge pressure of 250 kPa. **Determine the mole number and the mass of the helium in the balloon.**

Solution -2:



Use the universal gas constant, $R_u = 8.31447 \text{ kJ/kmol} \cdot \text{K}$

Apply ideal gas state equation to find mass:

$$N = \frac{PV}{R_u T} = \frac{351.3250 \text{ kPa} \times 523.988 \text{ m}^3}{8.31447 \text{ kJ/kmol} \cdot \text{K} \times 298.15 \text{ K}} = \mathbf{74.26 \text{ kmol}}$$

Mass

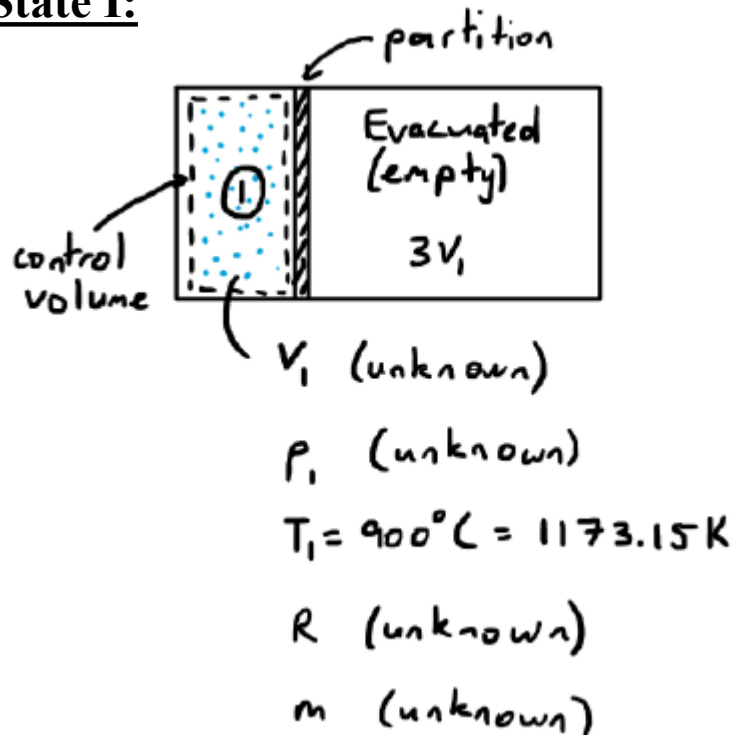
$$mR = (MN)R = NR_u \longrightarrow PV = NR_u T$$

Molar mass (M) x Mole number (N)

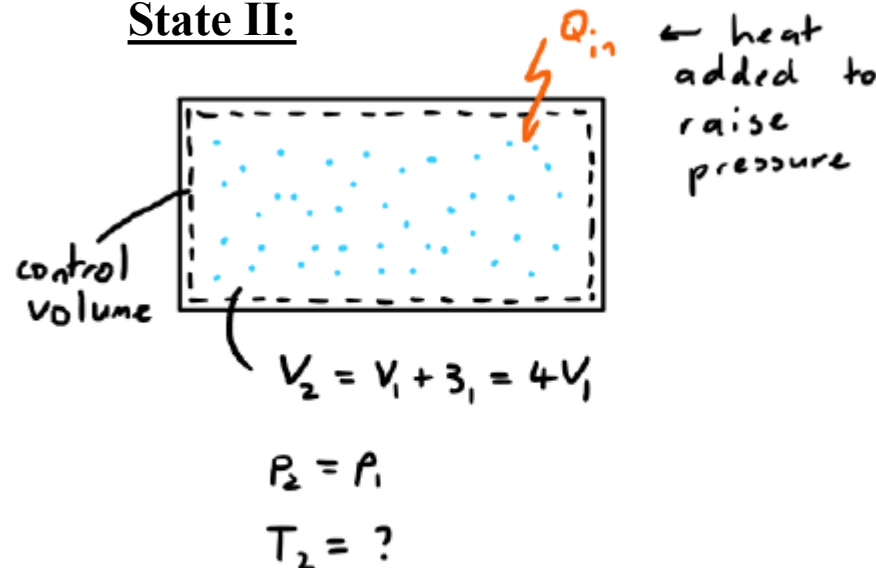
Q12. A rigid tank whose volume is unknown is divided into two parts by a partition. One side of the tank contains an ideal gas at 900°C . The other side is evacuated and has a volume three times the size of the part containing the gas. The partition is now removed, and the gas expands to fill the entire tank. Heat is now transferred to the gas until the pressure equals the initial pressure. **Determine the final temperature of the gas.**

Analysis:

State I:



State II:



Ideal gas equation

State 1: $P_1 V_1 = m R T_1$

State 2: $P_2 V_2 = m R T_2$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Ideal gas equation at two states for a fixed mass

Q12. A rigid tank whose volume is unknown is divided into two parts by a partition. One side of the tank contains an ideal gas at 900°C. The other side is evacuated and has a volume three times the size of the part containing the gas. The partition is now removed, and the gas expands to fill the entire tank. Heat is now transferred to the gas until the pressure equals the initial pressure. **Determine the final temperature of the gas.**

Solution:

Ideal gas equation

$$\text{State 1: } p_1 V_1 = m R T_1$$

$$\text{State 2: } p_2 V_2 = m R T_2$$



$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

Ideal gas equation at two states for a fixed mass

Re-arranging the equation to solve T_2 :

$$T_2 = \frac{T_1 p_2 V_2}{p_1 V_1}$$

Knowing $V_2 = 4V_1$ and $p_2 = p_1$

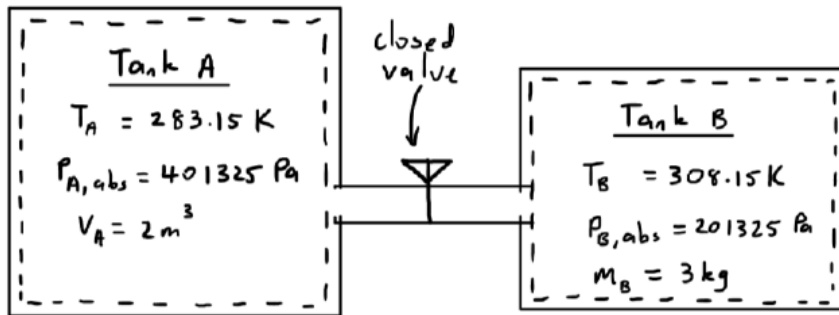
$$\Rightarrow T_2 = \frac{T_1 \cancel{p_2} V_2}{\cancel{p_1} V_1} = \frac{T_1 \cdot 4V_1}{V_1} = 4T_1$$

$$T_2 = 4 \times (900 + 273.15) = \mathbf{4692.60 \text{ K}}$$

Q13. A 2-m³ tank containing air at 10°C and 300 kPa is connected through a valve to another tank containing 3 kg of air at 35°C and 100 kPa. Now the valve is opened, and the entire system is allowed to reach thermal equilibrium with the surroundings, which are at 25°C. **Determine the volume of the second tank and the final equilibrium pressure of air. Give your answer in terms of absolute pressure.**

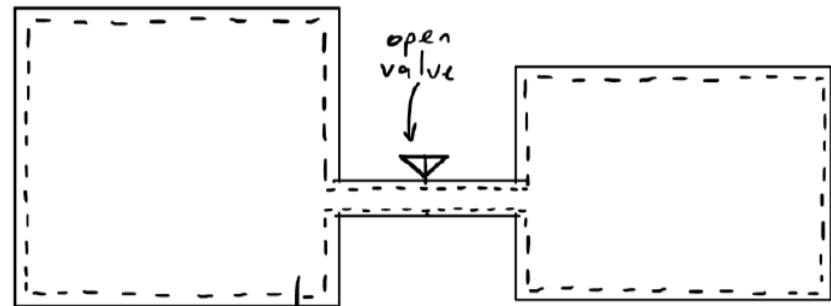
Analysis:

State I:



Before opening the valve: $V_B = \frac{m_B \cdot R_{air} \cdot T_B}{P_B}$

State II:



$T_2 = 25^\circ\text{C} = 298.15 \text{ K} \leftarrow \text{equilibrium temperature}$
 $V_2 = V_A + V_B = 2 + 1.3179 = 3.3179 \text{ m}^3 \leftarrow \text{total volume}$
 $R_{air} = 287 \text{ J/kg} \cdot \text{K}$

After opening the valve: **consider the two tanks as a single control volume system**

Q13. A 2-m³ tank containing air at 10°C and 300 kPa is connected through a valve to another tank containing 3 kg of air at 35°C and 100 kPa. Now the valve is opened, and the entire system is allowed to reach thermal equilibrium with the surroundings, which are at 25°C. **Determine the volume of the second tank and the final equilibrium pressure of air. Give your answer in terms of absolute pressure.**

Solution:

Before opening the valve: $V_B = \frac{m_B \cdot R_{\text{air}} \cdot T_B}{P_B} = \frac{3 \text{ kg} \times 0.2870 \text{ kJ/kg} \cdot \text{K} \times 308.15 \text{ K}}{(100 + 101.325) \text{ kPa}} = 1.3179 \text{ m}^3$

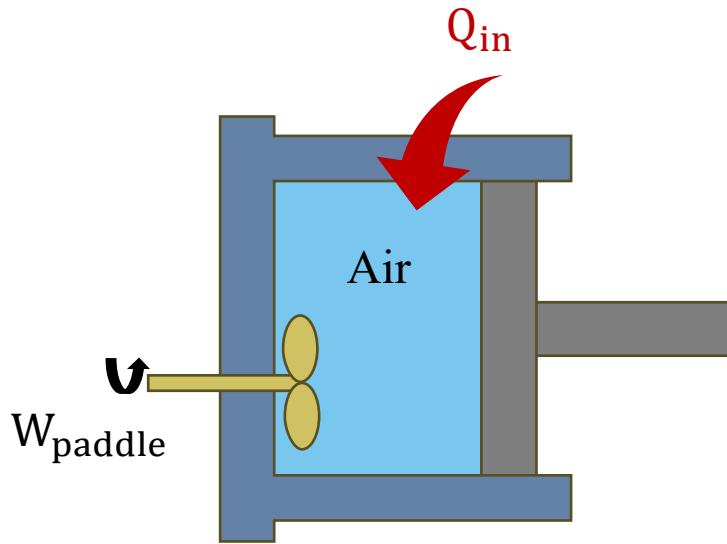
To find the total mass of air in the control volume system after opening the valve $m_{\text{total}} = m_A + m_B$

Find m_A at State I: $m_A = \frac{P_A V_A}{R_{\text{air}} T_A} = \frac{(300 + 101.325) \text{ kPa} \times 2 \text{ m}^3}{0.2870 \text{ kJ/kg} \cdot \text{K} \times 283.15 \text{ K}} = 9.8771 \text{ kg}$ then $m_{\text{total}} = 9.8771 + 3 = 12.8771 \text{ kg}$

Find the final pressure of air at State II: $P_{\text{II}} = \frac{m_{\text{total}} R_{\text{air}} T_2}{V_{\text{total}}} = \frac{12.8771 \text{ kg} \times 0.2870 \text{ kJ/kg} \cdot \text{K} \times 298.15 \text{ K}}{(1.3179 + 2) \text{ m}^3} = 332.1055 \text{ kPa}$

Q14. A piston-cylinder arrangement contains 0.04 m^3 of air at 50°C and 500 kPa . Heat is added in the amount of 60 kJ and work is done by a paddle wheel until the temperature reaches 600°C . If the pressure is held constant, **how much paddle-wheel work must be added to the air?**

Analysis: First Law of Thermodynamics for a closed system $\Rightarrow$ Energy balance



At system level

$$\Delta E = \Delta U + \Delta KE + \Delta PE \rightarrow \Delta E = \Delta U$$

$\swarrow$ $\swarrow$
 0 0

$$\Delta U = Q_{in} + W_{paddle} \Rightarrow W_{paddle} = \Delta U - Q_{in}$$

In this case, the change in internal energy is attributed by the temperature rise of air $\Delta U = mC_p\Delta T$

Q14. A piston-cylinder arrangement contains 0.04 m^3 of air at 50°C and 500 kPa . Heat is added in the amount of 60 kJ and work is done by a paddle wheel until the temperature reaches 600°C . If the pressure is held constant, **how much paddle-wheel work must be added to the air?**

Solution:

Find mass of air using ideal gas equation: $m = \frac{PV}{R_{\text{air}}T} = \frac{500 \text{ kPa} \times 0.04 \text{ m}^3}{0.2870 \text{ kJ/kg} \cdot \text{K} \times 323.15 \text{ K}} = 0.2156 \text{ kg}$

$$\Delta U = mC_p\Delta T = 0.2156 \text{ kg} \times 1.005 \frac{\text{kJ}}{\text{kg}} \cdot \text{K} \times [(600 + 273.15) - (50 + 273.15)]\text{K} = 119.1990 \text{ kJ}$$

$$W_{\text{paddle}} = \Delta U - Q_{\text{in}} = 199.1990 - 60 = \mathbf{59.1990 \text{ kJ}}$$